How to set up a CRISiSLab Raspberry Pi sensor

This is a modified version of our private internal document with IP addresses, IDs, etc removed. If you are a CRISiSLab member or partner, you may request access to this live internal version: How to set up a sensor (CRISiSLab internal)

This is a guide for configuring an individual sensor installation, for use in a system with the architecture outlined in "CRISiSLab Platform Documentation" (which should be co-located with this document. You may also check the source document: CRISiSLab Platform Documentation).

** indicates that the step applies to Raspberry Shake only
^^ indicates that the step applies to only devices that need an external pi to run the adapter program (i.e. not a Raspberry Shake)

1. Set up the physical sensor. Make sure the network cable is plugged in well and is connected to a router that will allow it to access the internet and that you can also access to control it.
2. **^^** Connect the sensor to the adapter pi
3. SSH into the Raspberry Pi
    a. ** **ssh myshake@rs.local**
       Password: **shakeme**
    b. ^^ **ssh pi@raspberrypi.local** (Note that the address might be something other than raspberrypi.local depending on the config. You might need to find the IP in your router)
       Password: **raspberry**
4. Install Zerotier **curl -s https://install.zerotier.com | sudo bash**
5. Connect to network **sudo zerotier-cli join ZEROTIER_NETWORK_ID_HERE**
    a. To get zerotier address **zerotier-cli status** (Send the output of this to Raj to get it approved. Then you can continue)
    b. To get MAC address `ifconfig`
    c. To get serial number **cat `/sys/firmware/devicetree/base/serial-number`**
6. ** Add datacast target in web ui:
    a. IP: **SERVER_IPV4_HERE**
    b. PORT: **2098 (or your different server UDP ingest port)**
7. Add sensor to admin.crisislab.org.nz (or your other instance of https://github.com/crisislab-platform/admin/ that is pointed at your server)
   Don't forget to add the zerotier IP!
    a. To get zerotier IP: **sudo zerotier-cli get ZEROTIER_NETWORK_ID_HERE ip**
    b. Also don't forget to add the host's name and contact email
    c. Add coordinates. If you're in the same place as the sensor, you can use the button to use your current location.
8. Make sure it is not a duplicate.

To get IP address of device connected directly via ethernet with no router:

1. Run **ping 192.168.1.255** for 5 seconds
2. Run **arp -a**
3. The ip next to *EN0* is the IP of the connected device